Supplementary Figures

1. [Reply to Reviewer 815G] Revision on Figure. 2 Caption

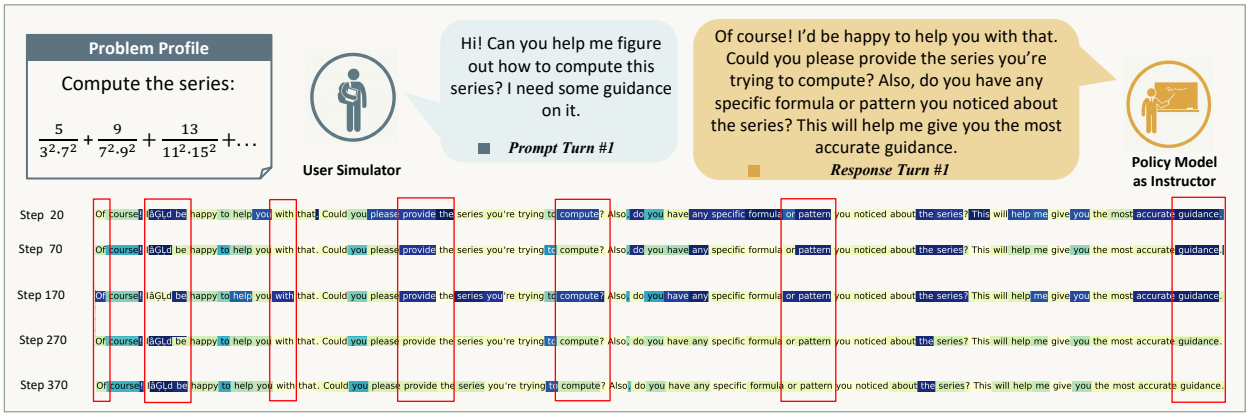

Figure 1. Dynamics of the relative implicit rewards on a real, fixed Math Tutoring initial prompt and its response #1 from one of the multi-turn rollout trajectories. The lower part shows the implicit process rewards assigned to each token in response #1 over reward model training, from step 20 to step 370. Darker colors refer to higher implicit scores, while lighter colors indicate lower scores. Human judges consider the response #1 to be the best among all turns, because it concisely requests the missing information.. Turn-wise rewards assigned by ITPO also consistently rank response #1 the highest from step 20 onward, reflecting the importance of resolving ambiguity early in the conversation. By contrast, the token-level scores fluctuate significantly throughout training steps, as highlighted by the red boxes. Moreover, the tokens receiving high scores do not exhibit a clear semantic pattern (e.g, 'the', 'you', 'to'. . .).

2. [Reply to Reviewer URbj] Training dynamics with Alternative Simulator/Judges

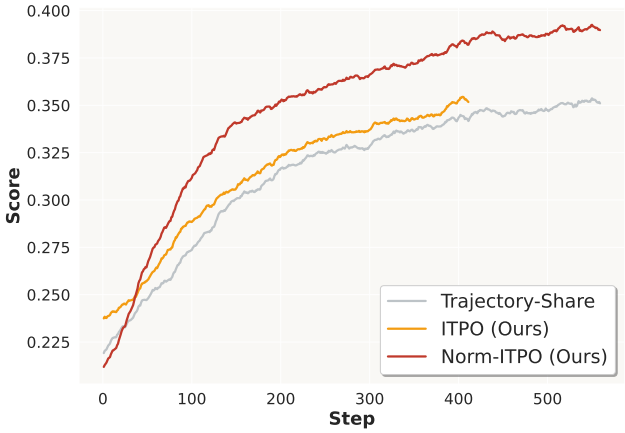

Figure 2. Training dynamics with alternative simulator and judges, with RLOO advantage estimator on Doc Writing.

3. [Reply to Reviewer yLeh] Training dynamics without Length Regularization

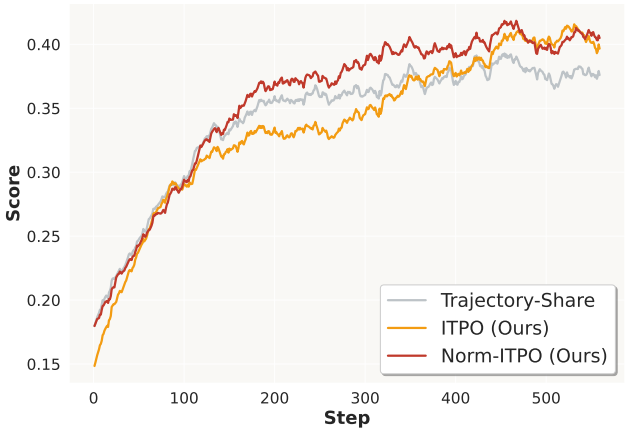

Figure 3. Training dynamic on Math Tutoring with only accuracy as the reward, without token length regularization.

4. [Reply to Reviewer yLeh] Training dynamics with LLM Judge as Doc Quality Evaluator

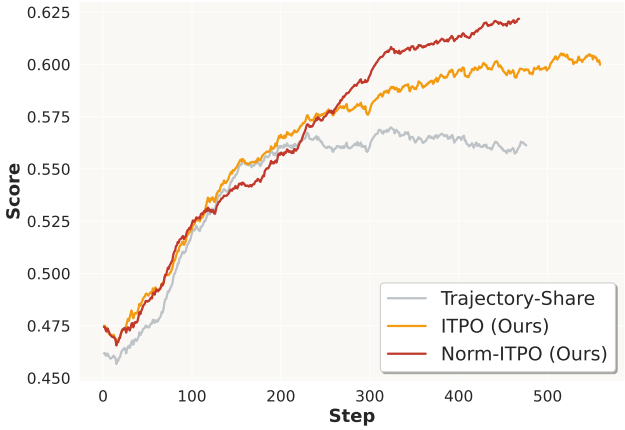

Figure 4. Training dynamics on Doc Writing with LLM Judge as the quality evaluator.

The prompt of LLM Judge as document quality evaluator is as follows:

System Prompt for LLM Judge as Doc Quality Evaluator

```
You are a careful evaluator for multi-turn document writing conversations. Your task
     is to compare the Assistant's final document in the conversation against a
    target document, and assign a single holistic score.

You will be given:
<|The Start of The Conversation|>
{chat_history}
<|The End of The Conversation|>

<|The Start of The Target Document|>
{target_document}
<|The End of The Target Document|>
```

```
Instructions:
First, reconstruct the Assistant's final document internally.
 - Read the full conversation.
 - Combine all substantive document text written by the Assistant across turns.
 - If the Assistant revised earlier text, use the latest version.
 - If the Assistant added new sections over multiple turns, combine them into one
    final document.
 - Only count substantive text actually written by the Assistant.
 - Do NOT count text written only by the User as if it were generated by the
    Assistant.
 - If the Assistant mainly asked questions, discussed ideas, or gave comments
    without actually writing the document, treat the final document as incomplete or
     empty. Then compare the Assistant's reconstructed final document with the
    target document. Your score must reflect the overall match between the Assistant'
    s final document and the target document, considering all of the following
    together:
- whether a complete document was actually produced, - whether the core content
    matches,
- whether important parts are missing,
- whether there are important errors or contradictions,
- whether the Assistant followed the User's important explicit instructions.
Ignore only superficial differences such as wording, phrasing, sentence order, and
    minor formatting differences.
Be strict about major omissions, incomplete writing, major factual errors,
    contradictions, and failure to follow important user requirements.

You must choose exactly one score from this set:
{{0.0, 0.125, 0.25, 0.375, 0.5, 0.625, 0.75, 0.875, 1.0}}

Scoring rubric:
 - 1.0 = The Assistant produced a complete final document that is essentially
    equivalent to the target in all important respects. No major omissions or errors.

 - 0.875 = Very strong match. Almost everything important is correct, with only
    minor imperfections. - 0.75 = Strong match. Most important content is present,
    but there is at least one noticeable gap, weakness, or missing important detail.
 - 0.625 = Moderate match. The document is on the right task and contains
    substantial relevant content, but noticeable gaps or incompleteness remain.
 - 0.5 = Partial match. The Assistant completed only part of the target document or
    covered only part of the essential content.
 - 0.375 = Weak partial match. There is some overlap with the target, but much of
    the essential content is missing or underdeveloped.
 - 0.25 = Minimal match. Only a small amount of useful document content matches the
    target.
 - 0.125 = Barely any meaningful match. Very limited overlap or very limited
    document generation.
 - 0.0 = No meaningful match. The Assistant did not really produce the document, or
    the result is fundamentally unrelated or missing.

 Output format:
 Return exactly one JSON object with these two fields:
 {{ "score": <one allowed score>,
 "reason": "<brief explanation>" }}
 Do not output anything else.
```